

Master Specification: Unified Master Architecture (UMA) & Recursive Symbiotic Logic System (RSLS)

Classification: Symmetric-Hyperbolic Einstein-Relaxation System

Core Mechanism: Regularized Singular Convex State-Space Boundary

I. Foundational State Space

The UMA is defined on a 4-dimensional pseudo-Riemannian manifold $(\mathcal{M}, g_{\mu\nu})$ with metric signature $(+, -, -, -)$. The system is governed by the state vector $\mathbf{\Psi} = (g_{\mu\nu}, U^\mu, M)$:

* **Metric Tensor $(g_{\mu\nu})$:** Dynamic Lorentzian fabric.

* **Information Flow (U^μ) :** Timelike vector congruence $(U^\mu U_\mu = 1)$.

* **Thermodynamic Memory (M) :** Causal scalar relaxation field (Israel–Stewart class), constrained to $M \in [0, M_{\max}]$.

II. The Variational Principle & Singular Barrier

The system's dynamics are derived from a local, symmetrizable action. The potential $V(M)$ —the **Holographic Wall**—is a strictly convex logarithmic potential that enforces the information capacity of the continuum:

The divergence of $V'(M)$ as $M \rightarrow M_{\max}$ acts as an infinite repulsive potential, ensuring the invariance of the domain and the coercivity of mathematical entropy.

III. Governing Equations of Motion

The evolution is described by three coupled, quasilinear partial differential equations:

1. The Einstein Sector (Geometric Reaction)

Where $T_{\mu\nu}$ includes the stress-energy of the flow, the memory field, and the dissipative exchange.

2. The Relaxation Sector (Thermodynamic Memory)

This forces M to integrate kinematic compression $(-\nabla_\mu U^\mu)$ over time, creating a **causal relaxation lag** between geometric forcing and informational response.

3. The Damped Flow Sector (Kinematics)

The right-hand side includes the **Constraint Damping current**, ensuring the timelike foliation remains a dynamical attractor.

IV. Emergent Scale (ℓ_*) and the Entropy Wall

The RSLS generates a macroscopic, mesh-independent length scale ℓ_* that regularizes geometric singularities. As $M \rightarrow M_{\max}$, the infinite convexity of the entropy Hessian pinches the eigenvalues of the flux Jacobian, preventing characteristic crossing and generating a **continuous free-boundary transition in BV space**. The invariant thickness is:

This replaces the singularity with a dissipative, finite-thickness **Saturated Information Attractor**.

V. Numerical Falsification Kernel (Phase A)

Verification requires a **Strict Hyperbolic Solver** (No Artificial Laplacian Viscosity):

* **Characteristic Upwinding:** Fluxes $F_{i+1/2}$ computed solely via HLL Riemann solver logic using signal speeds λ_{pm} .

* **Implicit Cattaneo Update:** Concurrent solution for entropy flux J_r to respect causal delay τ_J without stiffness instability:

VI. Observability & Spectral Stability

The wall behaves as a frequency-dependent, non-Markovian complex impedance boundary. It is stable if the relaxation dynamics $(V''(M))$ isotropize gravitational stress faster than geometric focusing amplifies shear:

The resulting **Wigner Time Delay** (Δt_{echo}) serves as the primary falsifiable astrophysical signature.

Status: **Definitive Master Specification Locked.**